

Multiple Nmap scanning techniques were performed against the authorized scanme.nmap.org testing host, including TCP port scanning, UDP port scanning, service/version detection, and Nmap Scripting Engine (NSE) scanning. The results were analyzed to identify accessible services and gather information about the target system.

The Nmap assessment was limited to scanme.nmap.org, an Nmap-provided target intended for security scanning demonstrations. The assessment was limited to the scanning and enumeration activities documented in this report.

Item	Configuration
Target	scanme.nmap.org
IPv4 address observed	45.33.32.156
Tool	Nmap 7.94
Scanner	Kali Linux
Testing activities	TCP scanning, UDP scanning, service enumeration, NSE scripts

Nmap scan configuration

Scan	Command	Purpose
TCP	nmap scanme.nmap.org	Identify commonly accessible TCP ports
UDP	nmap -sU scanme.nmap.org	Identify UDP services
Service detection	nmap -sV scanme.nmap.org	Identify service/version information
NSE scripts	nmap -sC scanme.nmap.org	Run Nmap's default scripts for additional enumeration

TCP Port Scanning

```

(root@kali)-[/home/kali]
# nmap scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:33 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.086s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
139/tcp    filtered   netbios-ssn
445/tcp    filtered   microsoft-ds
9929/tcp   open       nping-echo
31337/tcp  open       Elite

Nmap done: 1 IP address (1 host up) scanned in 4.15 seconds

```

The initial TCP scan identified multiple ports in different states. Ports 22/tcp and 80/tcp were identified as open and associated with SSH and HTTP respectively. Ports 9929/tcp and 31337/tcp were also identified as open. Ports 25/tcp, 139/tcp, and 445/tcp were reported as filtered, meaning Nmap could not establish that those ports were openly accessible.

UDP Port Scanning

```

(root@kali)-[/home/kali]
# nmap -sU scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:34 PDT
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.082s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 closed udp ports (port-unreach)
PORT      STATE      SERVICE
68/udp    open|filtered  dhcp
123/udp    open        ntp
135/udp    filtered     msrpc
136/udp    filtered     profile
137/udp    filtered     netbios-ns
138/udp    filtered     netbios-dgm
139/udp    filtered     netbios-ssn
445/udp    filtered     microsoft-ds
1900/udp   open|filtered  upnp

Nmap done: 1 IP address (1 host up) scanned in 1155.64 seconds

```

A UDP scan was performed using the “-sU” option. The scan identified UDP/123 as open and associated it with NTP. Several other ports were reported as filtered or open|filtered. UDP

scanning took significantly longer than the TCP scan, completing in approximately 1155 seconds. The UDP scan generated a retransmission warning, indicating that Nmap reached its retransmission limit for at least one port. This limitation was considered when interpreting the UDP results, particularly ports reported as “open|filtered”.

Service and Version Enumeration

```
(root@kali)~[/home/kali]
# nmap -sV scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:36 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.087s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.22 seconds
```

Service/version detection was performed using Nmap's “-sV” option. The scan identified OpenSSH on port 22 and Apache HTTP Server on port 80. Nmap also identified an Nping echo service on port 9929. Port 31337 was identified as “tcpwrapped”, indicating that Nmap could not determine a more specific service through this scan. The identified software versions provide information that can be used during subsequent vulnerability assessment and verification.

NSE Script Enumeration

```

(root@kali)-[/home/kali]
# nmap -sC scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:42 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.10s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open      http
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo
31337/tcp open      Elite

Nmap done: 1 IP address (1 host up) scanned in 7.33 seconds

```

The Nmap Scripting Engine was used with the “-sC” option to perform additional enumeration. The scripts returned SSH host-key information and HTTP metadata, including the HTTP title and favicon information. This additional information supplements the basic port and service enumeration performed by the previous scans.

Nessus Vulnerability scan

The screenshot displays the Tenable Nessus Essentials web interface. The main panel shows a table of scan results for the host `scanme.nmap.org`. The table has columns for Host, Auth, and Vulnerabilities. The scan status is 'Fail' with a score of 33. The right-hand panel provides details about the scan, including the policy, status, severity base, scanner, start/end times, and elapsed time. A donut chart at the bottom right shows the distribution of vulnerabilities by severity.

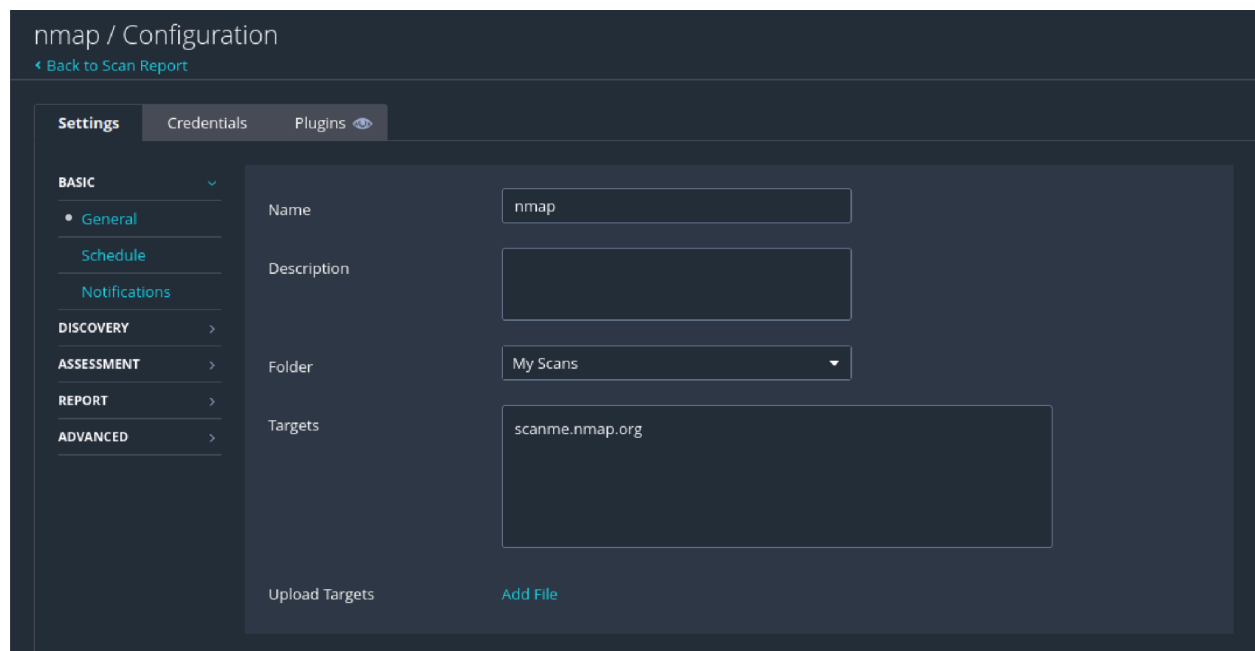
Host	Auth	Vulnerabilities
scanme.nmap.org	Fail	33

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 9:06 AM
- End: Today at 9:14 AM
- Elapsed: 8 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info



The Nessus scan was used to identify vulnerabilities and security weaknesses associated with the services discovered during Nmap enumeration.

The completed scan identified findings across multiple severity levels, including critical, medium, low, and informational findings.

False Positive Analysis and Verification

Finding verification methodology

Nessus findings were compared against Nmap service-enumeration results.

For example:

Nessus:

Port: 22/tcp

Service: SSH

Finding: SSH Terrapin Prefix Truncation Weakness

Nmap:

22/tcp open ssh

OpenSSH 6.6.1p1 Ubuntu

The Nmap result confirms that SSH is actually exposed on the target and provides additional service information.

False-positive procedure

For each important Nessus finding:

1. Confirm the affected port is open with Nmap.
2. Confirm the service identified by Nessus is actually running.
3. Compare the detected software/version with the Nessus finding.
4. Review the Nessus plugin description and evidence.
5. Do not claim exploitation unless an authorized validation test actually demonstrates it.
6. Record the result as **Verified**, **Requires Further Validation**, or **False Positive**.

Finding	Verification	Result
SSH Weak Algorithms	Nmap confirms SSH on 22/tcp; Nessus enumerated supported algorithms	Verified condition
CVE-2023-48795	Nmap confirms SSH service; Nessus reports Terrapin condition	Requires/has Nessus-based verification; exploitation not demonstrated

Verification: The Nessus finding was compared against the operating-system information identified during service enumeration. Nessus reported Ubuntu 14.04 and explicitly identified the system as unsupported. The finding was therefore treated as a confirmed vulnerability rather than an unverified detection.

MEDIUM

SSH Weak Algorithms Supported

Description

Nessus has detected that the remote SSH server is configured to use the Arcfour stream cipher or no cipher at all. RFC 4253 advises against using Arcfour due to an issue with weak keys.

Solution

Contact the vendor or consult product documentation to remove the weak ciphers.

See Also

<https://tools.ietf.org/html/rfc4253#section-6.3>

Output

The following weak server-to-client encryption algorithms are supported :

aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

The following weak client-to-server encryption algorithms are supported :

aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

To see debug logs, please visit individual host

Port 22

Hosts scanme.nmap.org

Plugin Details

Severity: Medium

ID: 90317

Version: 1.4

Type: remote

Family: Misc.

Published: April 4, 2016

Modified: January 12, 2026

Risk Information

Risk Factor: Medium

CVSS v2.0 Base Score: 4.3

CVSS v2.0 Vector: CVSS2#AV:N/AC:M/Au:N/CP:/L:N/A/N

MEDIUM

SSH Terrapin Prefix Truncation Weakness (CVE-2023-48795)

Description

The remote SSH server is vulnerable to a man-in-the-middle prefix truncation weakness known as Terrapin. This can allow a remote, man-in-the-middle attacker to bypass integrity checks and downgrade the connection's security.

Note that this plugin only checks for remote SSH servers that support either ChaCha20-Poly1305 or CBC with Encrypt-then-MAC and do not support the strict key exchange countermeasures. It does not check for vulnerable software versions.

Solution

Contact the vendor for an update with the strict key exchange countermeasures or disable the affected algorithms.

See Also

<https://terrapin-attack.com/>

Output

Supports following chacha20-poly1305 Client to Server algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

Supports following CBC Client to Server algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

Supports following Encrypt-then-MAC Client to Server algorithm : hmac-md5-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-ripemd160-etm@openssh.com,hmac-sha1-96-etm@openssh.com,hmac-md5-96-etm@openssh.com,hmac-md5@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-ripemd160,hmac-ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96

Supports following ChaCha20-Poly1305 Server to Client algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

more...

To see debug logs, please visit individual host

Port 22

Hosts scanme.nmap.org

Plugin Details

Severity: Medium

ID: 187315

Version: 1.6

Type: remote

Family: Misc.

Published: December 27, 2023

Modified: July 10, 2026

VPR Key Drivers

Threat Recency: No recorded events

Threat Intensity: Very Low

Exploit Code Maturity: PoC

Age of Vuln: 730 days +

Product Coverage: Very High

CVSSv3 Impact Score: 3.6

Threat Sources: No recorded events

Risk Information

Vulnerability Priority Rating (VPR): 5.0

Exploit Prediction Scoring System (EPSS): 0.9331

Risk Factor: Medium

CVSS v3.0 Base Score: 5.9

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:N/H:V/N

CVSS v3.0 Temporal Vector: CVSS:3.0/E:P/RL:C/RC:C

CVSS v3.0 Temporal Score: 5.3

CVSS v2.0 Base Score: 5.4

CVSS v2.0 Temporal Score: 4.2

Risk Information

Vulnerability Priority Rating (VPR): 5.0

Exploit Prediction Scoring System (EPSS): 0.9331

Risk Factor: Medium

CVSS v3.0 Base Score: 5.9

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:H/PR:N
/UI:N/S:U/C:N/I:H/A:N

CVSS v3.0 Temporal Vector: CVSS:3.0/E:P
/RL:O/RC:C

CVSS v3.0 Temporal Score: 5.3

CVSS v2.0 Base Score: 5.4

CVSS v2.0 Temporal Score: 4.2

CVSS v2.0 Vector: CVSS2#AV:N/AC:H/Au:N/C:N
/I:C/A:N

CVSS v2.0 Temporal Vector:
CVSS2#E:POC/RL:OF/RC:C

Vulnerability Information

CPE: cpe:/a:openbsd:openssh

Exploit Available: true

Exploit Ease: Exploits are available

Patch Pub Date: December 18, 2023

Vulnerability Pub Date: December 18, 2023

Reference Information

CVE: [CVE-2023-48795](#)

CRITICAL Canonical Ubuntu Linux SEoL (14.04.x)

Description

According to its version, Canonical Ubuntu Linux is 14.04.x. It is, therefore, no longer maintained by its vendor or provider.

Lack of support implies that no new security patches for the product will be released by the vendor. As a result, it may contain security vulnerabilities.

Solution

Upgrade to a version of Canonical Ubuntu Linux that is currently supported.

See Also

<http://www.nessus.org/u?5afc4199>

Output

OS : Ubuntu Linux 14.04

Security End of Life : April 25, 2019

Time since Security End of Life (Est.) : >= 7 years

To see debug logs, please visit individual host

Port ▲

Hosts

80 / tcp / www

[scanme.nmap.org](#)

Plugin Details

Severity: Critical
ID: 201408
Version: 1.2
Type: combined
Family: General
Published: July 3, 2024
Modified: March 26, 2025

Risk Information

Risk Factor: Critical
CVSS v3.0 Base Score: 10.0
CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H
CVSS v2.0 Base Score: 10.0
CVSS v2.0 Vector: CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

Vulnerability Information

CPE: cpe:/o:canonical:ubuntu_linux
Unsupported by vendor: true

Sev ▼

CVSS ▼

VPR ▼

EPSS ▼

Name ▲

Family ▲

Count ▼

⚙

☐	CRITICAL	10.0			Canonical Ubuntu Linux SEoL (14.04.x)	General	1	🔄	✎
☐	MIXED	...	...	...	SSH (Multiple Issues)	Misc.	6	🔄	✎
☐	MIXED	...	...	...	Openbsd Openssh (Multiple Issues)	Misc.	2	🔄	✎
☐	LOW	2.1 *	3.2	0.3159	ICMP Timestamp Request Remote Date Dis...	General	1	🔄	✎
☐	INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	3	🔄	✎
☐	INFO	...	...	...	Apache HTTP Server (Multiple Issues)	Web Servers	2	🔄	✎
☐	INFO	...	...	...	SSH (Multiple Issues)	General	2	🔄	✎
☐	INFO	...	...	...	SSH (Multiple Issues)	Service detection	2	🔄	✎
☐	INFO				Nessus SYN scanner	Port scanners	3	🔄	✎
☐	INFO				Service Detection	Service detection	3	🔄	✎
☐	INFO				Backported Security Patch Detection (WWW)	General	1	🔄	✎
☐	INFO				Common Platform Enumeration (CPE)	General	1	🔄	✎
☐	INFO				Device Type	General	1	🔄	✎
☐	INFO				Host Fully Qualified Domain Name (FQDN) ...	General	1	🔄	✎
☐	INFO				Nessus Scan Information	Settings	1	🔄	✎

Scan Details

Policy: Basic Network Scan

Status: Completed

Severity Base: CVSS v3.0 ✎

Scanner: Local Scanner

Start: September 21 at 9:06 AM

End: September 21 at 9:14 AM

Elapsed: 8 minutes

Vulnerabilities

Critical

High

Medium

Low

Info